MARTIN MOKOTLANA MAMADISE

Bachelor of Engineering Technology in Computer Engineering (Honours Student)

Bellville, Western Cape | martinmamadise80@gmail.com | LinkedIn | GitHub | Portfolio

PROFESSIONAL SUMMARY

Computer Engineering graduate and current Honours student with a strong foundation in software development, problem solving, and system design. Experienced in building and debugging applications using Java, C/C++, and Python, with exposure to backend development, embedded systems, and networked applications including IoT systems. Demonstrates the ability to design, develop, and troubleshoot real-world solutions, with a strong focus on writing efficient, maintainable code and continuously improving through hands-on projects. Seeking to contribute to a technology-driven organization in a graduate or entry-level role, where I can apply my skills to build reliable, scalable systems while growing as a software and systems engineer.

CORE TECHNICAL SKILLS

Programming Languages:
- Java (OOP, application development)
- C/C++ (systems programming, embedded development)
- Python (scripting, data handling)

Software Engineering:
- Object-Oriented Programming (OOP)
- Data Structures & Algorithms (foundations)
- Debugging & Problem Solving
- Software Development Life Cycle (SDLC)

Backend & Tools:
- REST API fundamentals
- SQL (MySQL - querying & data validation)
- Git (version control - project tracking & collaboration)

Systems & Networking:
- Networking Fundamentals (OSI model, routing, IP addressing)
- Network simulation (GNS3, VyOS)

Embedded & IoT:
- Arduino, ESP32 development
- Sensor integration & real-time data systems

PROJECT EXPERIENCE

Traffic Engineering using Graph Theory and Linear Programming (Python / Optimization)
- Designed and developed a network optimization system to improve traffic flow in large-scale computer networks
- Implemented Dijkstra's shortest-path algorithm as a baseline for routing efficiency analysis
- Formulated and solved linear programming models using PuLP and CBC Solver
- Developed multi-commodity flow (MCF) optimization to distribute traffic across multiple paths and reduce congestion
- Simulated real-world conditions using the NSFNET backbone topology
- Reduced maximum link utilisation from over 600% to below 75%, significantly improving scalability and performance
- Visualised network performance and utilisation using Matplotlib

Smart Agriculture Monitoring System (IoT + Embedded Systems)
- Designed and developed an IoT-based system to monitor environmental conditions such as soil moisture, temperature, and humidity
- Implemented ESP32-based data collection and communication using HTTP protocols
- Integrated multiple sensors to support real-time decision-making for irrigation
- Built and debugged system logic to ensure accurate data readings and system reliability
- Analysed system behaviour and improved performance through iterative testing and refinement

Brick Breaker WebGL Game (JavaScript / Graphics Programming)
- Developed an interactive browser-based game using WebGL and JavaScript
- Implemented collision detection algorithms and game physics logic
- Designed gameplay mechanics and handled rendering logic for real-time interaction
- Debugged performance and logic issues to ensure smooth gameplay experience

Network Virtualization & Routing Simulations (GNS3 / VyOS)
- Designed and configured **virtual network topologies** using GNS3
- Implemented routing protocols and verified **end-to-end connectivity**
- Analysed and resolved network issues through systematic debugging
- Simulated real-world networking scenarios to understand system behaviour under different conditions

TOOLS AND TECHNOLOGIES

- **Languages:** Java, JavaScript, C/C++, Python
- **Libraries & Frameworks:** NetworkX, Pandas, Matplotlib, PuLP
- **Version Control:** Git, GitHub
- **Databases:** MySQL
- **Development Tools:** VS Code, Apache NetBeans, CodeBlocks
- **Systems & Platforms:** Linux, VirtualBox
- **Networking Tools:** Wireshark, Cisco Packet Tracer, GNS3, VyOS
- **API & Testing:** Postman

- **Embedded Systems:** Arduino IDE, ESP32
- **Other:** MATLAB (basic), CBC Solver

EDUCATION

Bachelor of Engineering Technology in Computer Engineering (Honours) (2026 – in progress)
Cape Peninsula University of Technology

Bachelor of Engineering Technology in Computer Engineering (2023 – 2026)
Cape Peninsula University of Technology

National Senior Certificate (2022)
Kopa Secondary School